

Lab 11 - Rotational Inertia

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Learning Outcomes Objectives

- Determine how the rotational inertia of a system depends on how mass is distributed about the rotation axis
- Measure angular acceleration and use it to determine rotational inertia
- Analyze how changing one variable affects the motion of a rotating system
- Compare experimental results with a theoretical model

Pre-lab Principles

Definition: rotational inertia is resistance to change of rotation.

Statement: the more rotational inertia an object has, the harder it is to rotate.

1. Write a similar definition and statement for what you know about **inertia** from chapter 4.

Definition: **The tendency for an object to remain at rest or in motion, is related to its mass. Newton's First Law.**

Statement: **The more mass, the more inertia, the harder it is to change its motion**

The **rotational inertia** of a body depends on the body's mass and the way its mass is distributed about an axis of rotation. In general, the more compact an object, the less rotational inertia it has and the easier it is to rotate.

2. Answer If the mass of an object is spread out from its axis of rotation, will it be easier or harder to rotate? Does your answer agree with the first statement above?

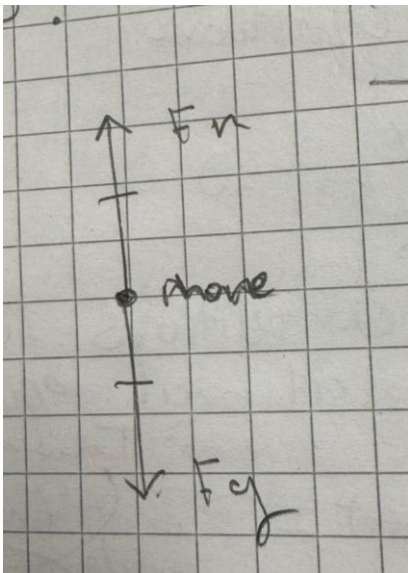
If mass is spread out from the axis of rotation, it will be harder to rotate. This agrees with the statement above "the more compact an object, the less rotational inertia it has and the easier it is to rotate".

3. Imagine sitting on a stool that rotates holding 2 dumbbells, one in each hand. A labmate starts you spinning as you hold the dumbbells close to your chest. Will you speed up or slow down as you straighten your arms and move the dumbbells away from your chest? Explain.

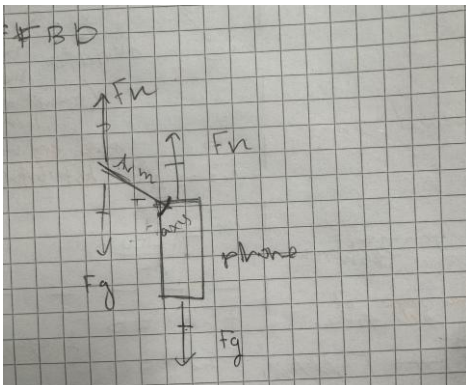
As you move the dumbbells away from your chest then you will slow down. This is because angular momentum is conserved but the radius is larger causing it to slow down.

In Lab 10, you found **rotational equilibrium** by hanging masses at different positions to balance a pole (meter stick) on its axis of rotation. (We did? Yes, you did!). Each mass hanging a distance away from the axis of rotation exerted a torque $\tau = r \times F$. You also learned that in equilibrium, the net torque on the system must be zero (in addition to net force = 0).

4. Draw a sketch of a force applied to an object at a distance away from an axis of rotation. For example, your hand exerts a normal force on your phone holding it at waist level above the ground. **Label the phone, the axis of rotation, the location and direction of the normal force F_n and the distance r from the axis (also called the length of the lever arm).**



5. Now make an Extended FBD of the situation including **both of the forces** acting on the phone while you hold it at waist level above the ground. **Use equations to show that both conditions of equilibrium are satisfied in rotational equilibrium!**



Torque Part 2 (Newton's second Law for rotation)

Newton's 2nd Law also has a rotational equivalent, similar to $F_{net} = ma$:

$$\tau_{net} = \sum_{i=1}^n \tau_i = I_{total} \alpha$$

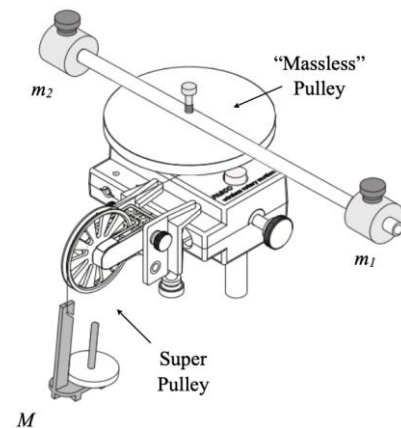
where I is the rotational inertia (also called "moment of inertia") and α the angular acceleration from chapter 6. **You will use this equation to experimentally determine the moment of inertia of a rod with 2 masses by varying the distance the masses are away from the axis of rotation.**

6. Determine the SI units of rotational inertia I .

KG and m $\cup$ "

Apparatus

1. bar and "point" masses
2. super pulley
3. wireless rotary motion sensor
4. calipers and meter stick
5. balance scale
6. support rod and base
7. mass hanger
8. $\sim 1.5\text{m}$ string



A hollow, cylindrical rod of radius r_{rod} , length L , and mass m_{rod} is free to rotate about an axis through its central diameter.

Two discrete masses m_1 and m_2 are placed at distances of r_1 and r_2 from the rotation axis. The rod is attached to a "massless" pulley of radius R through the rod's axis of rotation.

7. **Label r_1 and r_2 in the figure shown. Draw the string** following the grooves of both pulleys.

This "massless" pulley is connected by a very light string to a mass M that hangs over a frictionless "super" pulley. M , via the string, exerts a perpendicular torque τ when it is allowed to fall from rest, which causes the massless pulley to rotate and have an angular acceleration α .

8. **Rewrite** the equation for torque τ in terms of F and R . Identify F and R .

9. **Draw** a free-body diagram (FBD) of the hanging mass M . Side view only.

10. **Draw** an extended free-body diagram (EFBD) of the rod and masses, where $m_1 = m_2 = m$ and $r_1 = r_2 = r$. **Side view and Top view** are needed to identify the perpendicular torque τ !

Part 1: Setup and Measurements

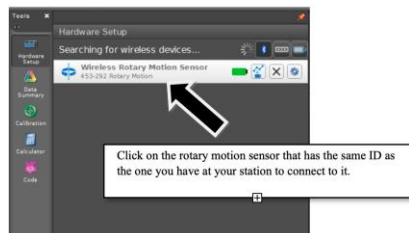
1. **Measure** and **record in googlesheet**:

- Mass of the rod m_{rod}
- Length of the rod L
- Mass m (the same for m_1 and m_2)
- Mass M of the hanging mass (hanger)

2. **Measure** the diameter to the largest pulley groove

3. **Determine** the radius R from the axis of rotation to the largest pulley groove.

4. **Start** Capstone, turn on the rotary motion sensor, and click on hardware setup.



5. **Open** a new graph and use the “add new plot area” icon to **add a second plot area that shares the same time axis**.

Set the top graph to record the angular velocity ω of the massless pulley in rad/s.

Set the bottom graph to record the position of the hanging mass M in m.

Part 2: Varying Mass Position

You will perform **multiple trials** by changing the distance r of the masses m from the center.

Important:

- The two masses m must be at equal distances r from the center on opposite sides of the rod. This is how you ensure that $m_1 = m_2 = m$ and $r_1 = r_2 = r$
- The masses m and M will remain the same for all trials
- The radius R of the massless pulley groove is also constant
- The only variable that changes is r

Experimental trials: record in googlesheet

- Trial 1: small r (close to center)
- Trial 2: medium r
- Trial 3: larger r
- Trial 4: largest safe r (mass flush with end of rod)

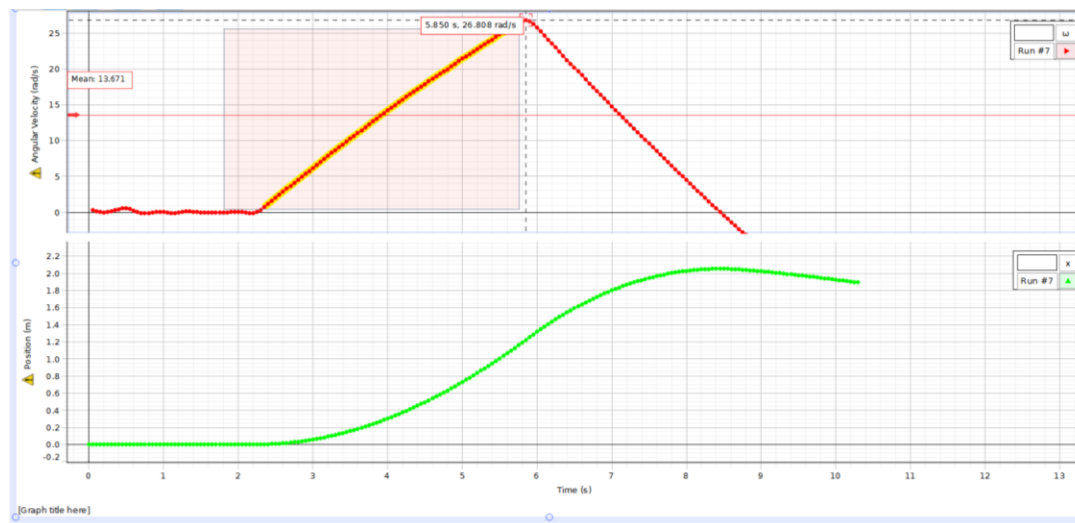
For each trial:

1. Measure and record r (use calipers from center-to-center of thumb screws)
2. Wind the string around the large groove so the hanging mass is near the top.
3. Start recording.
4. Release the mass from rest.
5. Stop recording **before** the mass hits the ground.
6. Determine the angular acceleration α from the angular velocity ω vs. time graph.
Explain how you obtained α using one of the Capstone graphs. (Hint: Lab 7).

Part 3: Data analysis

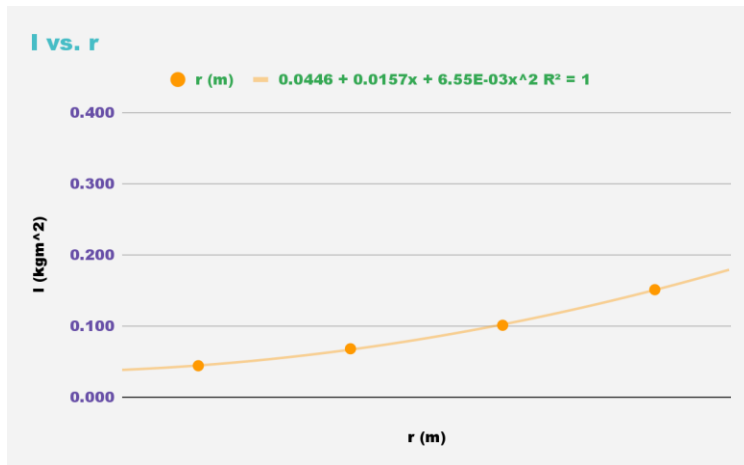
1. **Use** a google sheet to create a table including the following measurements for each trial:

- r the distance of the masses from the center rotation axis
- α the angular acceleration of the rotating disc, rod and masses
- a the linear acceleration of the hanging mass
- F_{net} the net force on the hanging mass
- T the tension in the string
- τ_{net} the net torque applied to the rotating disc by the hanging mass
- I the moment of inertia of the rod and masses



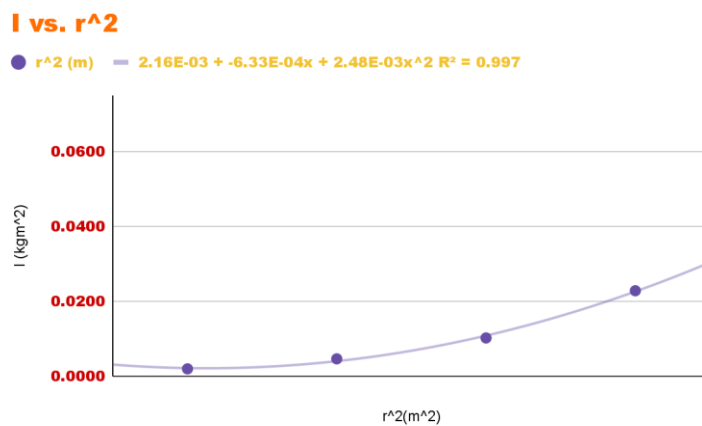
	m_rod	L_rod	m masses	M hanger	d pulley	R pulley					
measured units	27.8	38.1 cm	75.4 g	49.5	52.4mm	2.1 cm					
SI units	0.0278	0.381	0.0754	0.0495	0.0524	0.021					
				W = Mg							
Trial Measurements											
Trial #	r (cm)	r (m)	$\Delta\omega$ (rad/s)	Δt (s)	α (rad/s ²)	a (m/s ²)	F_net (N)	T (N)	τ _net (Nm)	I (kgm ²)	r ² (m)
1	4.4	0.044	16.164	5.25	3.1	0.065	0.00320	0.482	0.0101	0.16	0.0020
2	6.8	0.068	13.671	5.85	2.3	0.049	0.00243	0.483	0.0101	0.21	0.0046
3	10.1	0.101	11.117	7.32	1.5	0.032	0.00158	0.484	0.0102	3.14	0.0102
4	15.1	0.151	8.154	7.85	1.0	0.022	0.00108	0.485	0.0102	0.47	0.0228
							M_hanger (a)	M_hanger (g - a)	R_pully (T)	T / α	

2. **Use** the relationship between linear and angular acceleration ($a = \alpha R$) to calculate the linear acceleration **a** of the hanging mass for each trial.
3. **Use** Newton's 2nd law to calculate the net force **F_{net}** on the hanging mass **M** for each trial.
4. **Calculate** the tension **T** in the string from **F_{net}** on the hanging mass and **F_g**.
5. **Calculate** the net torque **τ_{net}** applied to the massless pulley by combining the tension and the position where it is applied for each trial.
6. **Use** Newton's 2nd law for rotating objects to determine the moment of inertia **I** of each trial.
7. **Create** a graph of **I** vs. **r**. What trendline is a best fit to the data?



A linear trend best fits the data on the graph.

8. **Create** a graph of I vs. r^2 . What trendline is a best fit to the data?



The best line that fits the data would be the exponential graph

Reflection Questions

1. How does moving mass farther away from the rotation axis affect rotational inertia?

Moving farther away will increase an object's rotational inertia, because the inertia depends on the mass x square of the distance of the axis, so moving out/farther it increases the object resistance in the rotational speed.

2. How does this change the motion of the system?

Moving the mass further away from the axis increase the rotational inertia, which causes the system to resist the change of the rotation by accelerating slowly and the rotation is more steadily under the same torque that is applied

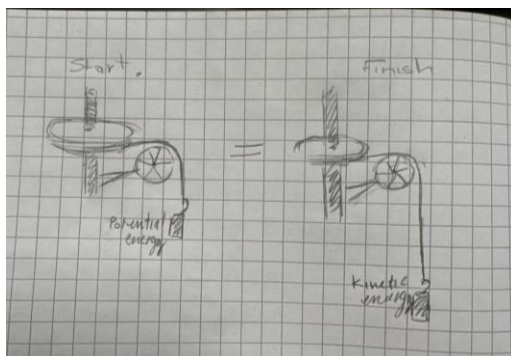
3. Based on your graphs, what relationship best describes how rotational inertia depends on distance from the axis?

The relationship between rotational inertia is proportional to the square of the distance (r^2).

The graph of I vs. r shows a linear line of best fit, confirming that as the masses move farther from the axis of rotation, the resistance to rotation increases.

4. How could this lab be expanded to test conservation of energy? Include a sketch in your answer.

We can see the conversion of energy from potential to kinetic energy as the hanging mass falls. We can also observe the loss of kinetic energy through the friction of the pulley system. If we compare the $Mgh = \frac{1}{2}(m)(v^2) + \frac{1}{2}(I)(\omega^2)$



Sample calc for Fnet

$$F_{net} = m_{hanger}(a)$$

$$F_{net} = 0.0495kg(0.065\frac{m}{s^2}) = 0.0032175N$$

Sample calc for T

$$T = m_{hanger}(g - a)$$

$$T = 0.0495kg(9.81\frac{m}{s^2} - 0.065\frac{m}{s^2}) = 0.4823775N$$

Sample calc for τ_{net}

$$\tau_{net} = R_{pulley}(T)$$

$$\tau_{net} = 0.021m(0.482N) = 0.010122Nm$$

Sample calc for I

$$I = \frac{T}{\alpha}$$

$$I = \frac{0.482N}{3.1} = 0.15548Kg(m)^2$$

Setup Measurements

	m_rod	L_rod	m masses	M hanger	d pulley	R pulley
measured units	27.8	38.1 cm	75.4 g	49.5	52.4mm	2.1 cm
SI units	0.0278	0.381	0.0754	0.0495	0.0524	0.021

$W = Mg$

Trial Measurements

Trial #	r (cm)	r (m)	$\Delta\omega$ (rad/s)	Δt (s)	(rad/s ²)	a (m/s ²)	F_net (N)	T (N)	_net (Nm)	I (kgm ²)	r ² (m)
1	4.4	0.044	16.164	5.25	3.1	0.065	0.00320	0.482	0.0101	0.16	0.0020
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M_hanger (a)	M_hanger (g)	R_pully (T)	T / (α)
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range of distance r for trials 1 - 4			
r1	r2	r3	r4
4.0 - 4.5 cm	8.5 - 9.0 cm	13.0 - 13.5 cm	17.5 - 18.0 cm

